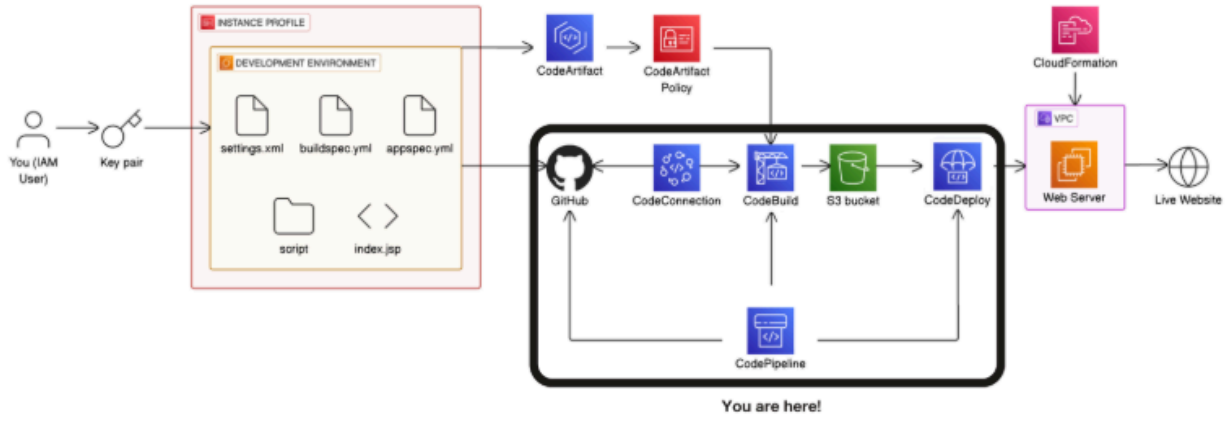
AWS CI/CD Pipeline Pt4



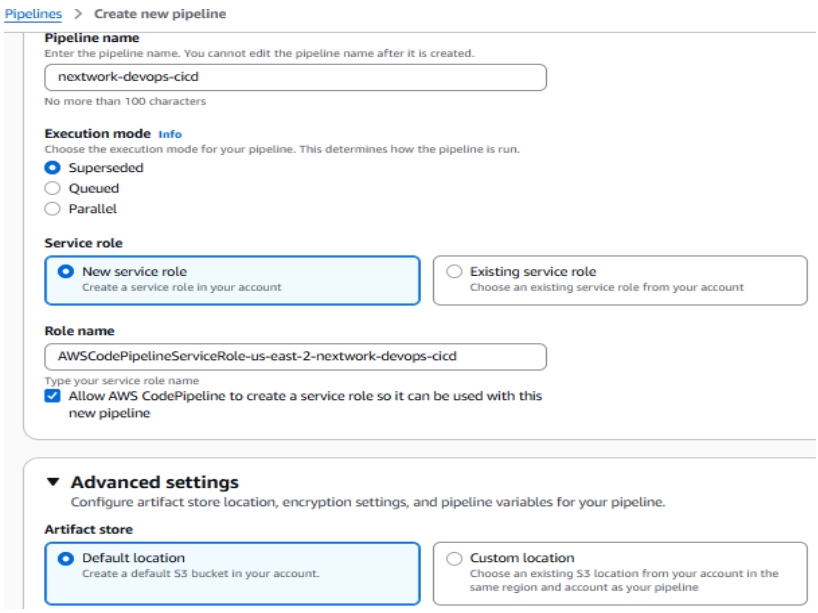
Set up your pipeline

Start a new pipeline

- Head to the CodePipeline console
- Select the **Getting started** page. Welcome to CodePipeline!
- In the CodePipeline dashboard, Select **Create pipeline**.
- Select **Build custom pipeline**

Configure Pipeline Settings

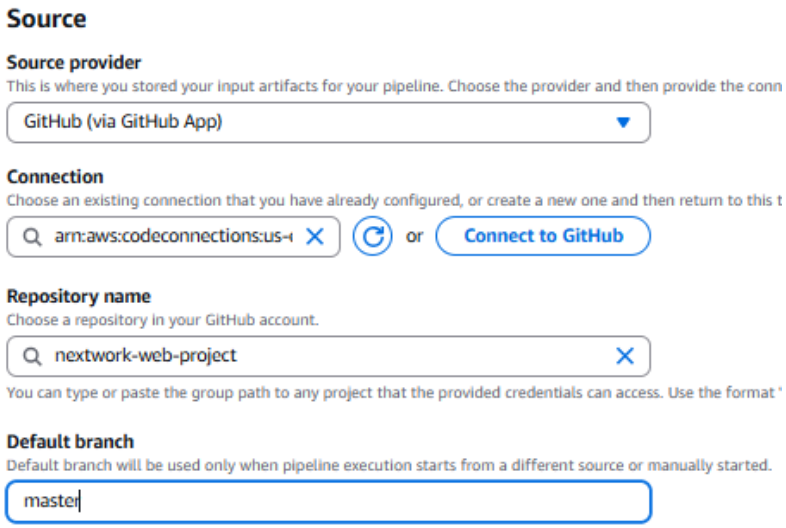
- Name your pipeline **nextwork-devops-cicd**
- Under **Execution mode**, select **Superseded**
- Under **Service role**, select **New service role**. Keep the default role name.
- Expand **Advanced settings**.
- Leave the default settings for **Artifact store**, **Encryption key**, and **Variables**
- Click **Next**

- 

Source Stage

Now, let's configure the Source stage of our pipeline. This is where we'll tell CodePipeline where to fetch our source code from.

- In the **Source provider** dropdown, select **GitHub (via GitHub App)**
- Under **Connection,** select your **existing GitHub connection**
- Under **Output artifact format,** leave it as **CodePipeline default.**



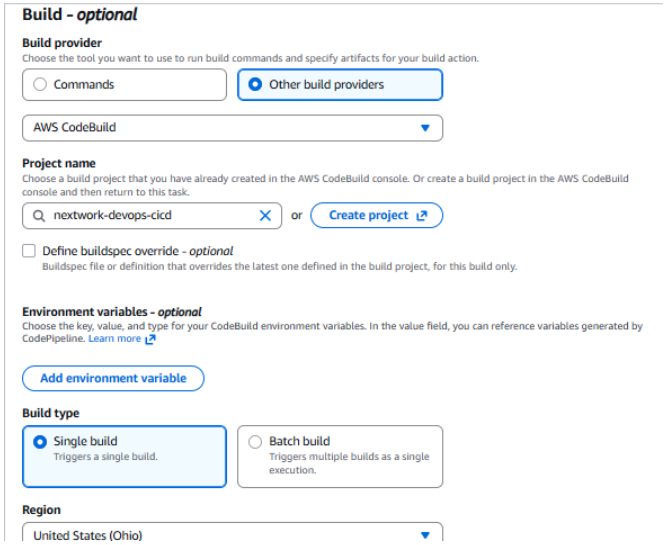
- Make sure that **Webhook events** is checked under **Detect change events**

Build stage

The Build stage is where our source code gets transformed into a deployable build artifact.

We'll tell CodePipeline to use AWS CodeBuild to compile and package our web application.

- In the **Build provider** dropdown**,** select **AWS CodeBuild from Other build providers.**
- Under **Project name,** select your existing CodeBuild project
- In the Project name dropdown, search for and select **nextwork-devops-cicd**
- Leave the default settings for **Environment variables, Build type, and Region**
- Under **Input artifacts, SourceArtifact** should be selected by default.



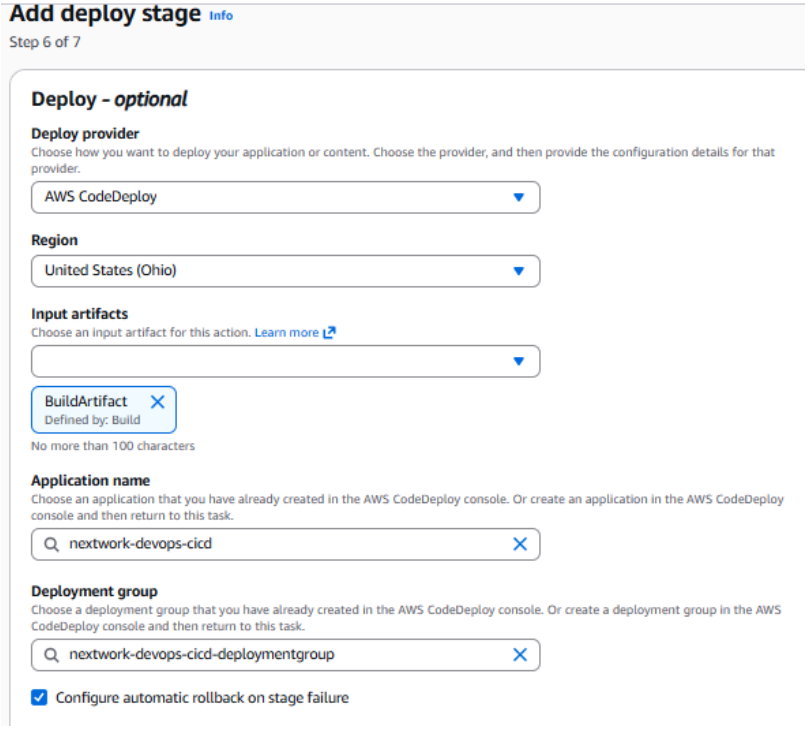
- Click **Next.**

Skip Test Stage

- On the **Add test stage** page, click **Skip test stage**

Deploy Stage

- In the **Deploy provider** dropdown, select **AWS CodeDeploy**.
- Under **Input artifacts**, **BuildArtifact** should be selected by default.
- Under **Application name**, select your existing CodeDeploy application
- Under **Deployment group**, select your existing CodeDeploy deployment group
- Check the box for **Configure automatic rollback on stage failure**



Run Your Pipeline!

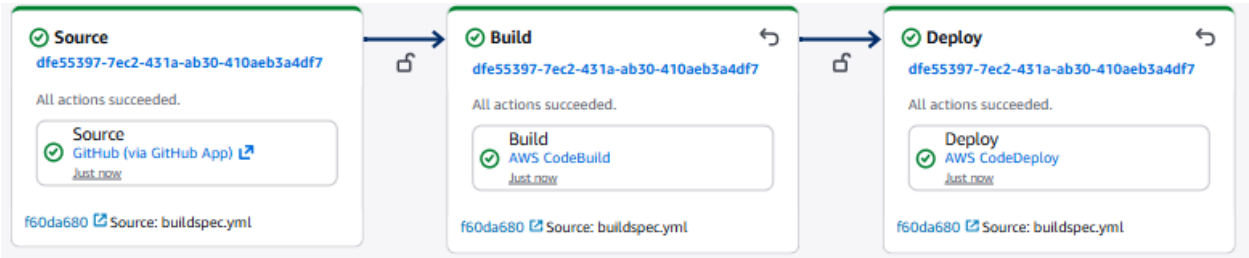
Let's watch our pipeline run for the first time! This will help us verify that everything is working correctly.

Review Your Pipeline

- On the **Review** page, take a moment to review all the settings you've configured for your pipeline.

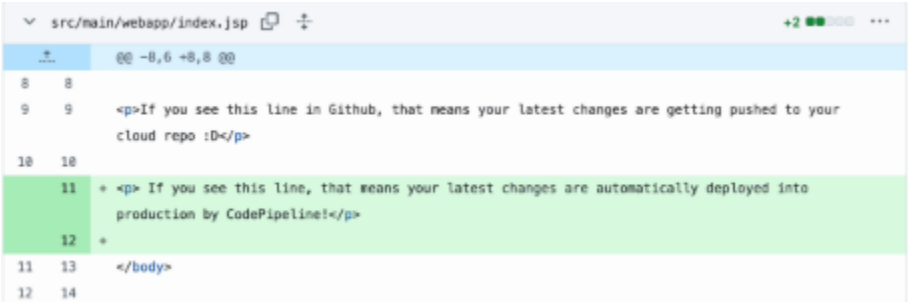
Run Your Pipeline

- After clicking **Create pipeline**, you will be taken to the pipeline details page.
- Note the pipeline diagram at the top of the page.

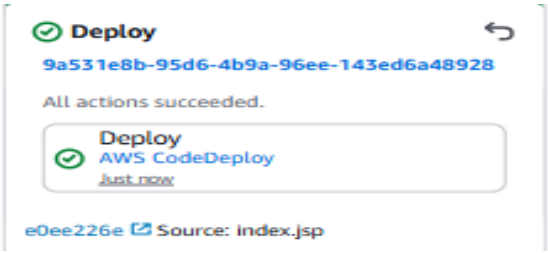


Test Pipeline with Code Change

- Open your web app code in your local IDE (e.g., VS Code).
- Open the index.jsp file located in src/main/webapp/.
- Add a new line in the <body> section of index.jsp
    - If you see this line, that means your latest changes are automatically deployed into production by CodePipeline!
- Save the index.jsp file.
- Open your terminal and navigate to your local git repository for the web app.
- Commit and push the changes to your GitHub repository using the following commands:
    - git add .
    - git commit -m "Update index.jsp with a new line to test CodePipeline"
    - git push origin master
- Hint: If having issues with Github credentials run this:
    - git config --global --unset credential.helper
    - git config --local --unset credential.helper
    - git remote set-url origin https://<github_username>@github.com/<github_username>/<repository_name>.git
- You should see a new execution starting automatically after you push the changes to GitHub
- Click on the **Source** stage box in the pipeline diagram.
- Scroll down in the stage details panel to see the commit message
- Click on the **Commit ID** link in the Source stage details panel
- This should open the commit page in your GitHub repository in a new browser tab.
- Verify that the commit page shows the code changes you just pushed (the new line you added to index.jsp

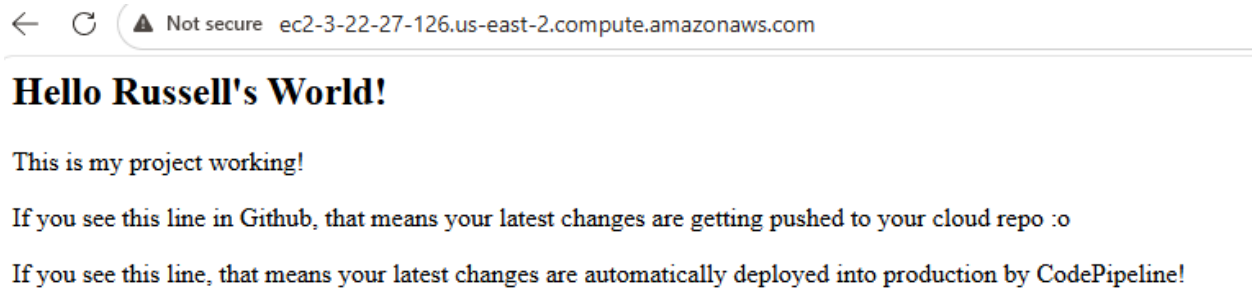


Wait for the **Build** and **Deploy** stages to complete successfully (turn green) in the CodePipeline console.



Verify Automated Deployment

Let's try accessing the web app to see your code change live!

- To find the Public IPv4 DNS, in the CodePipeline console, click on the **Deploy** stage, then click on the **CodeDeploy** link in the details panel.
- In the CodeDeploy console, scroll down to **Deployment lifecycle events** and click on the **Instance ID**.
- On the EC2 instance summary page, copy the **Public IPv4 DNS**

WOOOOOO 😭🙏 This confirms that your latest code changes were automatically deployed by CodePipeline.